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Image Classification Task

Question and answer

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1. Long Question

1.1 Unsupervised Learning at eSewa

As a Data Scientist at eSewa, two high-impact areas where unsupervised learning can drive business value are customer segmentation and fraud pattern detection.

Customer segmentation involves understanding user behavior across eSewa's diverse customer base. The platform records a wide variety of user activities including mobile top-ups, utility bill payments, P2P transfers, and merchant transactions. Given this diversity, clustering algorithms such as K-Means or DBSCAN can be employed to group users based on features like transaction frequency, volume, transaction categories, and region. The business benefit is clear: segment-specific promotions can increase engagement, reduce churn, and boost transaction volume. For example, frequent utility payers may receive cashback offers for service renewals, while inactive users can be re-targeted with re-

engagement campaigns.

Fraud detection is another critical area. While supervised models require labeled fraud data, unsupervised approaches like Isolation Forest or Autoencoders are more effective in detecting previously unknown patterns. These models analyze transaction behaviors—such as unusual time of day, device ID switches, or location mismatches—to flag anomalies. Once integrated with real-time transaction systems, suspicious activities can be reviewed or blocked instantly. This not only reduces financial loss but also builds customer trust.

These models should be monitored regularly and retrained using live data streams. Their outputs can be visualized on internal dashboards, triggering workflows such as fraud investigation or automated marketing. In summary, unsupervised learning at eSewa enhances both user personalization and platform security, directly contributing to revenue growth and operational efficiency.

1.2 ML Engineering Challenges in E-Commerce

In real-world e-commerce environments, machine learning systems face challenges not only during model development but also post-deployment. Three key challenges are data drift, imbalanced data, and serving latency.

Data drift occurs when user behavior or business conditions change over time, causing model input distributions to shift. This leads to performance degradation. For example, during festival seasons, spending habits shift, affecting recommendation relevance. If not addressed, this results in poor user engagement. To solve this, teams can implement monitoring systems to track feature distributions and trigger automatic retraining workflows using tools like TFX or Airflow.

Imbalanced data is another issue, especially in fraud detection or rare event modeling. If most training examples are from one class, the model may ignore the minority class. This can be dangerous if fraudulent transactions are underrepresented. Techniques like SMOTE, undersampling, or modifying class weights can mitigate the imbalance. Periodic validation using F1-score rather than just accuracy is essential.

System latency also plays a vital role in user experience. A delay in showing recommendations or confirming a payment can lead to user drop-off. Solutions include model optimization (quantization, pruning), low-latency infrastructure (e.g., ONNX, TF Serving), and caching frequent queries.

Finally, cross-functional collaboration ensures models are aligned with product goals. Data scientists, engineers, and business teams must co-design metrics, validate edge cases, and ensure that insights translate into measurable business outcomes.

2. Short Questions

2.1 Overfitting – Answer: Overfitting vs Underfitting

In machine learning, overfitting and underfitting describe how well a model generalizes to unseen data. Overfitting occurs when the model learns not only the patterns but also the noise in the training data. It performs exceptionally well on training data but poorly on new data due to lack of generalization. Underfitting, conversely, happens when the model is too simplistic to capture the underlying relationships, resulting in poor performance on both training and test sets.

For example, a decision tree that grows too deep may memorize every training example (overfitting), while a shallow linear regression might miss complex patterns in housing price prediction (underfitting). Both scenarios are problematic: overfitting increases variance, while underfitting increases bias.

To address overfitting, one might use regularization techniques like L1/L2 penalties, dropout in deep networks, or reduce model complexity. For underfitting, increasing model complexity or using more expressive features can help. The ideal model balances bias and variance, providing good performance on unseen data.

2.2 Neural Network Architecture – Answer: CNN vs RNN

Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) serve different purposes in deep learning. CNNs are specialized for spatial data such as images, using convolutional filters to capture local patterns like edges and textures. RNNs, in contrast, are designed for sequential data such as time series or text, with memory units that retain information from previous time steps.

For instance, CNNs are widely used in medical image classification or facial recognition, whereas RNNs are suitable for language modeling, chatbot responses, or stock price forecasting. The main difference lies in data structure: CNNs process fixed-size grids, while RNNs handle variable-length sequences.

Both networks face training challenges. CNNs can overfit when trained on small datasets. This is mitigated using techniques like dropout or data augmentation. RNNs suffer from vanishing gradients when sequences are long, hindering learning. Solutions include using LSTM or GRU cells that preserve gradients over time.

Other solutions include batch normalization to stabilize learning and early stopping to prevent overfitting. Understanding the nature of the data helps determine whether a CNN, RNN, or a hybrid model is most appropriate.

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